

From Local to Cloud-Native: Real-World Azure Arc in Action

Azure Fest 2026 – September 23rd

Tom Claes



```
1
2 {
3   "name": "Tom Claes",
4   "age": 27,
5   "current_role": {
6     "title": "Cloud Solution Architect",
7     "company": "Microsoft",
8     "experience_years": 4
9   },
10  "education": {
11    "institution": "KU Leuven",
12    "program": "Master of Science in Artificial Intelligence",
13    "duration_years": 6
14  },
15  "community_contribution": {
16    "focus": "Sharing knowledge and contributing to the community",
17    "platforms": [
18      "Microsoft Tech Community",
19      "Medium",
20      "GitHub",
21      "Public speaking engagements"
22    ]
23  }
24 }
```

First things first, thank you to all sponsors!



Agenda



Why Azure Arc?



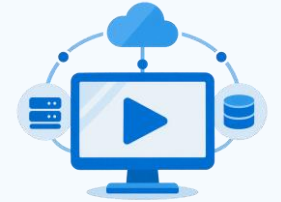
What is Azure
Arc?



What is Azure Arc
not?



How to get
started?



Demo

Why Azure Arc?

Raise of hands

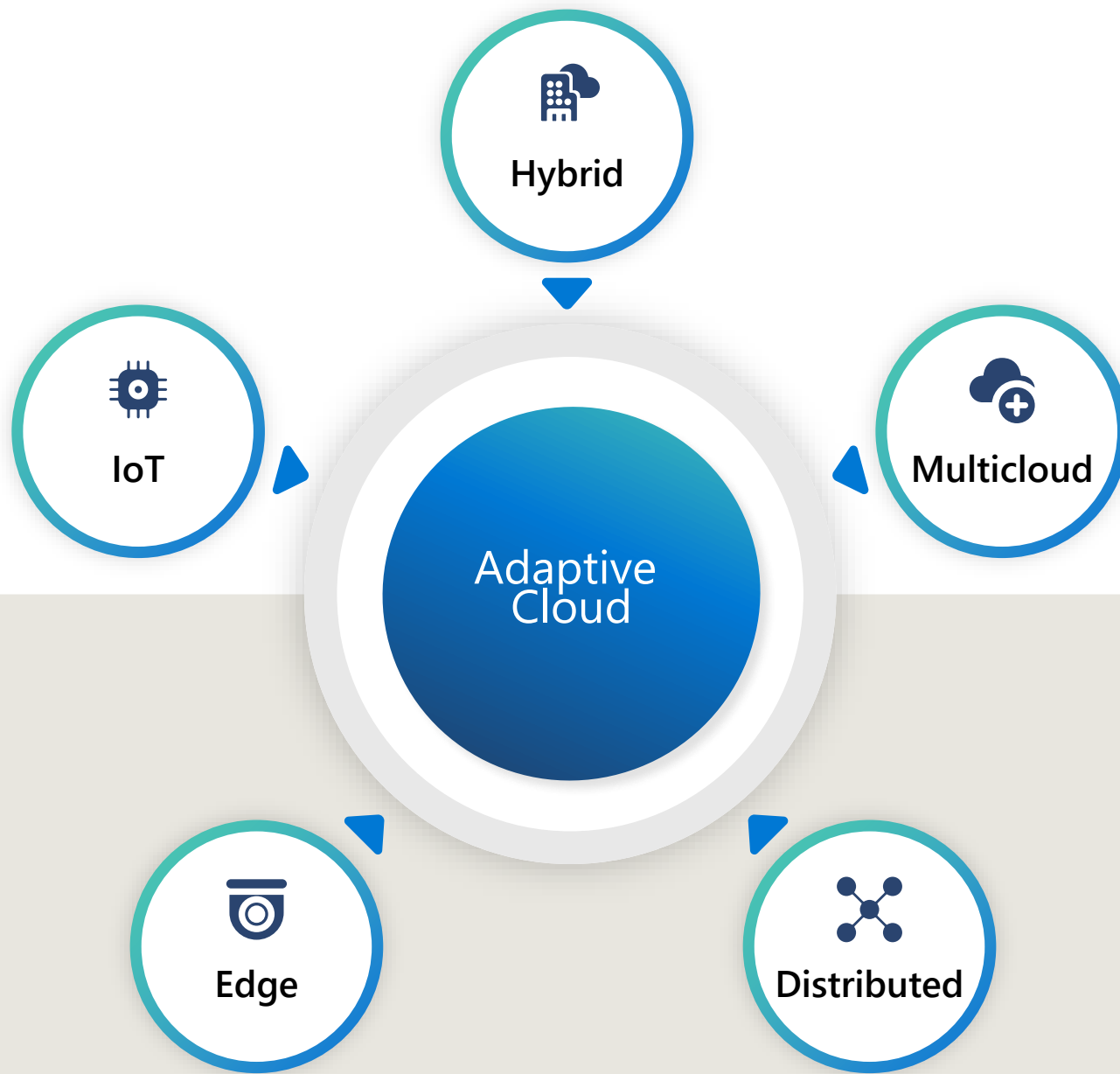
- Who is not familiar with Azure? Who knows about Azure Arc?
- Are you operating in a hybrid environment?
Azure, multi-cloud and/or on-prem?
- Inventory: how many servers (Windows/Linux/SQL) does your organization operate? How many k8s clusters?

Now I will stop bothering you

- EOS: How to identify servers reaching end of support (e.g., SQL Server 2016 & Windows Server 2016)?
- Patching: Can I get one view showing update compliance across the organization?
- Monitoring: When something goes wrong, how many monitoring tools does your operations team need to open?
- Sovereignty: how can I keep my data & servers on-prem while benefiting from cloud capabilities?

Why Azure Arc?

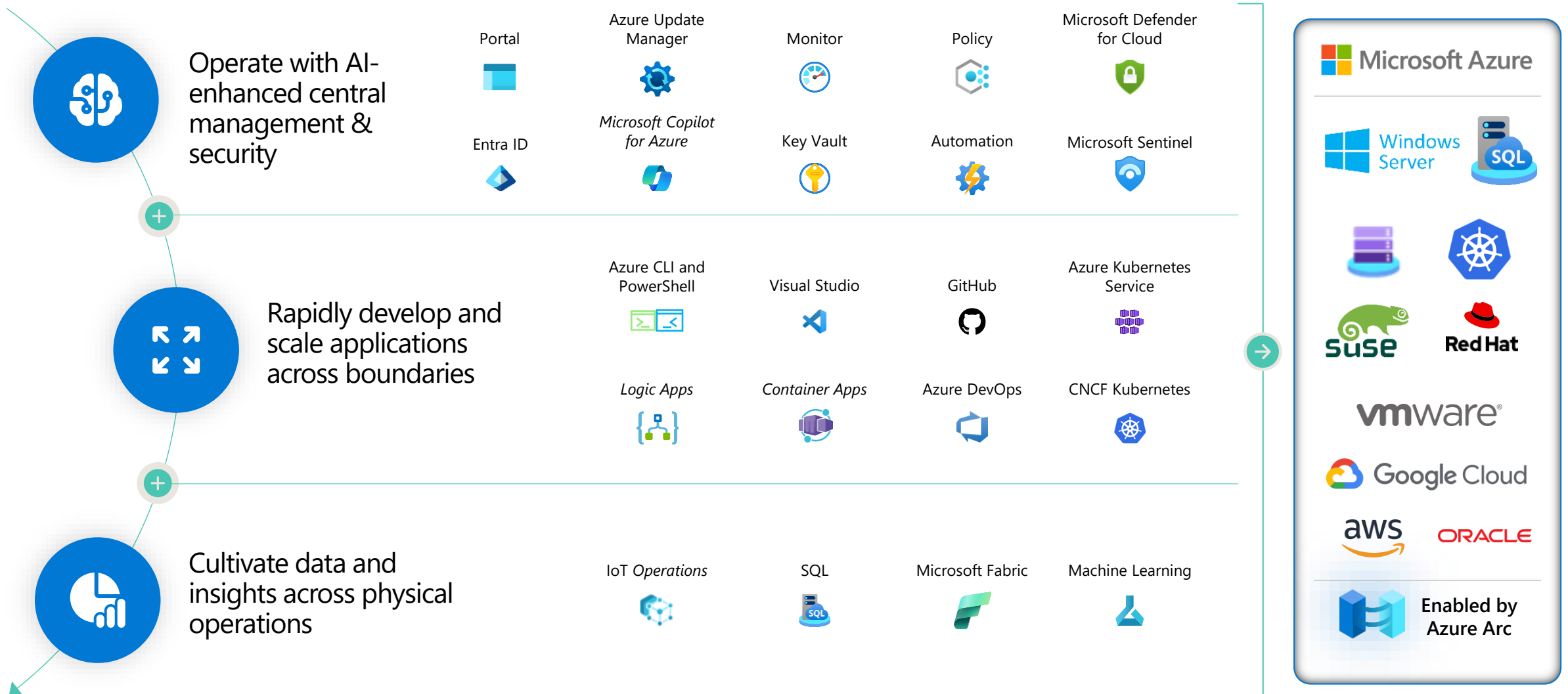
1. Hybrid & multicloud estate under one control plane
2. Unified view & insights across on-prem, edge and other clouds
3. Standardized tooling – patching, monitoring, desired state configuration
4. Modernization of legacy management stacks – e.g., SCCM, WSUS
5. Governance at scale – Azure Policy, RBAC, tags, Defender
6. Cost savings – fewer 3rd-party licenses, Software Assurance benefits, SQL PAYG via Arc



Advancing ~~Hybrid~~ cloud to ~~cloud~~ Adaptive cloud

Thrive in dynamic environments by unifying teams, sites, and systems across hybrid, multicloud, edge, and IoT.

Enabling adaptive cloud with Azure Arc



Microsoft Azure



Single control plane with Azure Arc

Infrastructure

Connect and operate
hybrid resources as native
Azure resources

Azure Arc-enabled infrastructure



Arc
Server



K8s



Windows



SQL
Server



Linux

Services

Deploy and run Azure services
outside of Azure while still
operating it from Azure

Azure Arc-enabled services



Microsoft
Defender



Azure
Monitor



Microsoft
Sentinel



Azure
Update
Manager



Azure
Policy



Multi-cloud



Datacenter



Edge

So, what is Azure Arc?

One Azure control plane – where management of every connected resource is coordinated



**Azure Resource
Manager**



**Resource
groups & tags**



Azure RBAC



Azure Policy



**Monitoring,
inventory &
updates**

What is Azure Arc not?

- Not a virtualization platform – VMware, Hyper-V keep running
- Not a one-way trip to Azure – workloads can stay where they are
- Not a platform replacement – Windows stays Windows, SQL stays SQL, K3s stays K3s
- Azure Arc is not a hardware
- Not an all or nothing enablement – choose what makes sense for your organization
- Azure Arc is not all you need: Azure Monitor, Azure Policy, AUM... (it's an enabler)
- Azure Arc is not expensive
- Azure Arc is not secretly capturing your personal data.

How to get started?

How does the onboarding work?

To install the agent package:

- Windows: PowerShell
 - Linux: Bash script
- Generated through the Azure Portal
- Built-in (by default) in WS2025 & SQL Server 2025

At scale deployments:

- PowerShell and Service Principal
- SCCM (task sequence or script)
- Ansible
- Windows Admin Center
- Group Policy
- SCVMM

Microsoft Azure

Home > Azure Arc | Machines

Onboard existing machines with Azure Arc

1. Select deployment method

Deployment method

- ☒ Basic script
- ☐ Configuration Manager
- ☐ Group Policy
- ☐ Ansible

2. Download the script and add service principal credentials

ⓘ You will need to manually add your service principal client ID and secret to the onboarding script. Verify that your service principal has the 'Azure Connected Machine Onboarding' role.
[Learn more about adding servers with a service principal](#)
[Create a service principal or copy an existing client ID](#)

```
1 $global:scriptPath = $myInvocation.mycommand.definition
2
3 function Restart-AsAdmin {
4     $pshCommand = "powershell"
5     if ($PSVersionTable.PSVersion.Major -ge 6) {
6         $pshCommand = "pwsh"
7     }
8
9     try {
10        Write-Host "This script requires administrator permissions to install the Azure Connected Machine Agent"
11        $arguments = "-NoExit -Command '& '$scriptPath'"
12        Start-Process $pshCommand -Verb runAs -ArgumentList $arguments
13        exit 0
14    } catch {
15        throw "Failed to elevate permissions. Please run this script as Administrator."
16    }
17 }
18
19 try {
20     if (-not ([Security.Principal.WindowsPrincipal] [Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([Security.Principal.WindowsBuiltInRole]::Administrator)) {
21         if ([System.Environment]::UserInteractive) {
22             Restart-AsAdmin
23         } else {
24             throw "This script requires administrator permissions to install the Azure Connected Machine Agent"
25         }
26     }
27 }
28
29 # Add the service principal application ID and secret here
30 $ServicePrincipalId="359947e8-fe13-483c-b58a-30833789447c";
31 $ServicePrincipalClientSecret="<ENTER SECRET HERE>";
```

Download

What is the Azure Arc Connected Machine Agent?

- Windows agent is distributed as a Windows Installer package (MSI)
- Linux: the preferred package format for the distribution (.rpm or .deb), hosted in the Microsoft package repository

Central agent with extensions management in the Portal

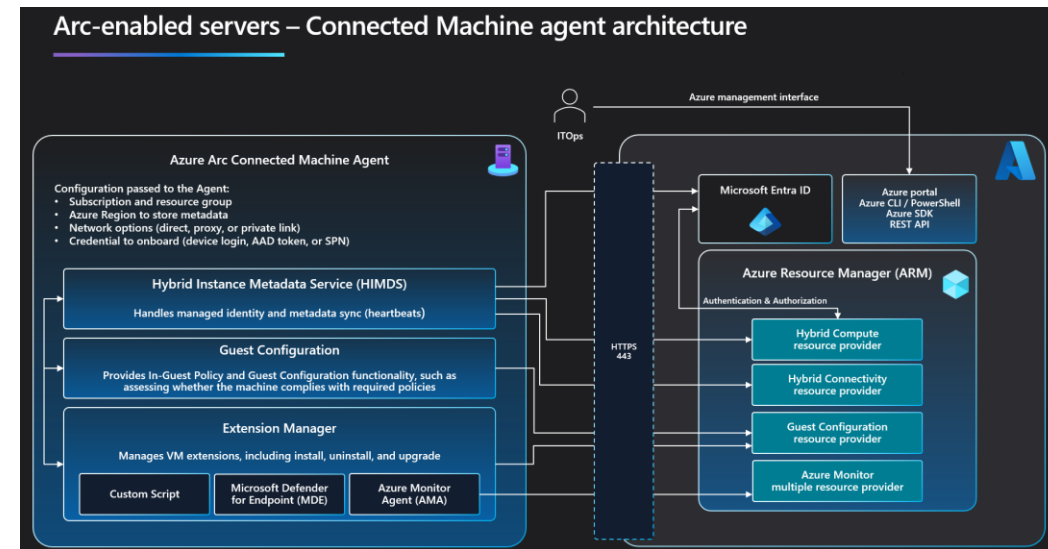
Resource utilization on the server

- Overview of the Azure Connected Machine agent - Azure Arc | Microsoft Learn

Metadata captured (depends on the extension as well)

- Overview of the Azure Connected Machine agent - Azure Arc | Microsoft Learn

Management: Manage and maintain the Azure Connected Machine agent - Azure Arc | Microsoft Learn



azcmagent CLI

Goal: configure, manage, and troubleshoot a server's connection with Azure Arc.

Important features:

- Tear-0 assets (e.g., domain controllers): [Security overview - Azure Arc | Microsoft Learn](#)
 - Disabling remote access capabilities
 - Setting an extension allowlist for the extensions you intend to use, or disabling the extension manager if you are not using extensions
 - Disabling the machine configuration agent if you don't intend to use machine configuration policies
- Allow/blocklist for unwanted extensions
- Connectivity configuration

More information: [azcmagent CLI reference - Azure Arc | Microsoft Learn](#)

Demo

Small Arc environment

- Azure Arc-enabled Servers (Linux & Windows)
- Azure Arc-enabled SQL Server
- Azure Arc-enabled k3s cluster (AKS EE)
- Extensions and features
- Connectivity to servers
- azcmagent CLI

Onboarding of a Windows Server in real-time

Thank you!

Any questions?



LinkedIn



Slides